

Course: B. Tech in Civil Engineering

Sem: IV

Subject : Structural Mechanics I (BTCVC403)

Max Marks:60

Date:30/11/2019

Duration: 3 Hr.

Instructions to the Students:

1. Solve ANY FIVE questions out of the following.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.
4. Solve All Questions in Sequential order.

Q. 1 Solve the following.

Marks

12

A) Find Static and Kinematic Indeterminacy of the structure as shown in Fig No 1

06

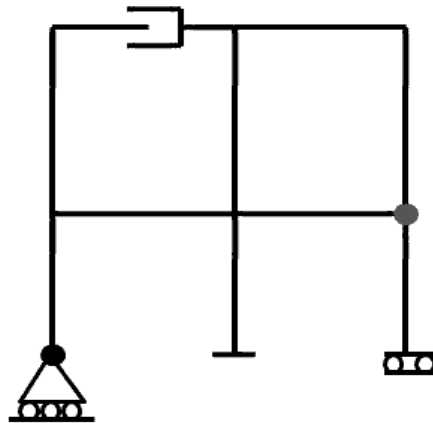


Fig No 1

B) Find Static and Kinematic Indeterminacy of the structure as shown in Fig No 2

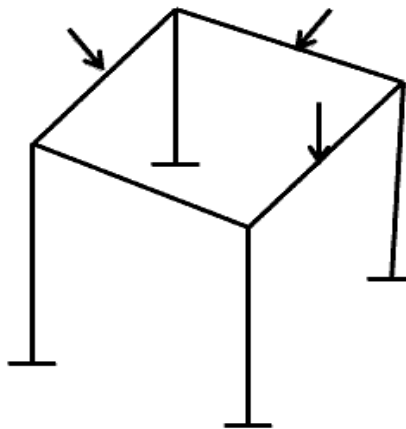


Fig No 2

06

Q.2 Solve the following.

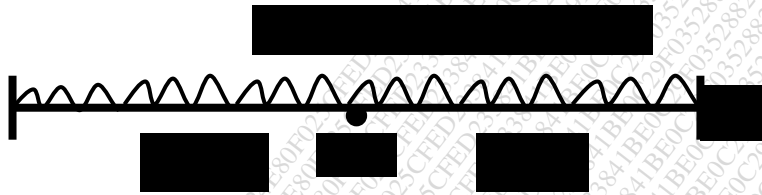
12

- A) A Beam AB is Simply Supported over a span 5m in length. A Concentrated load of 30kN is acting at section 1.25m from support A .Calculate the deflection under the point load .Also calculate the deflection at the center of the beam and Slope at end A for the beam. Take $E = 200,000 \text{ MPa}$ and $I = 13 \times 10^{-6} \text{ m}^4$.

Q. 3 Solve the following.

12

- A) Draw the Bending Moment diagram, Elastic curve, location of Point of contraflexure and maximum deflection for a Fixed beam subjected to UDL 5kN/m over a span of Length 10m. EI is Constant

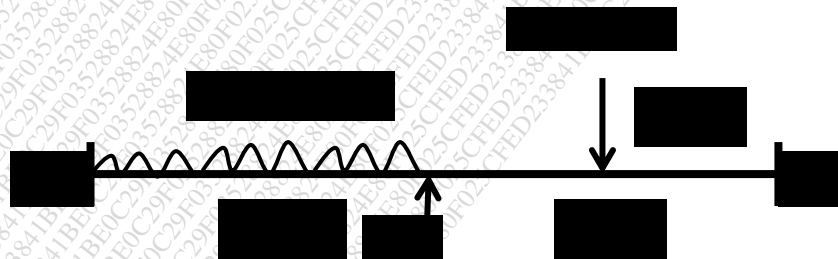


Q.4 Solve the following.

12

A Continuous Beam ABC is Supporteded at A, B, C. Span AB= 6m, Span BC = 5m. Span

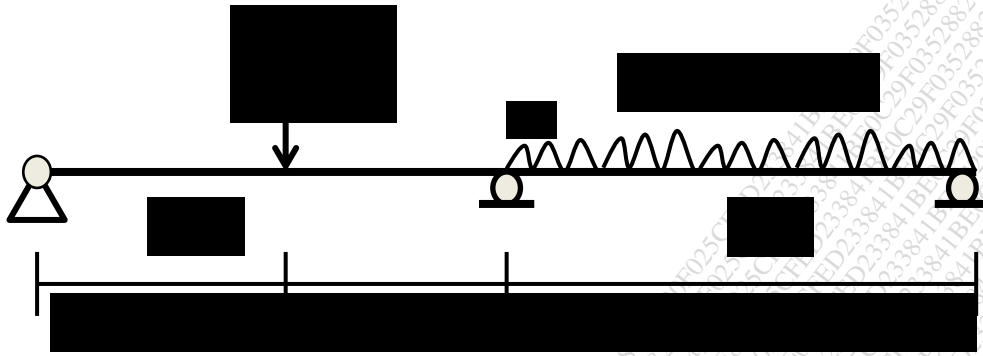
- A) AB carries a UDL of 30kN/m and Span BC Carries a UDL of 25kN/m. Calculate the Support moments and draw SFD, BMD by Hardy Cross Method. 06
- B) Draw BMD and SFD for the Beam as Shown in Figure. Also Find the reactions at Support. Use theorem of Three moments. 06



Q. 5 Solve the following.

12

- A) Determine the Support moments for the continuous beam as shown in fig below. The Relative values of MI are as shown in fig below. Use Slope Deflection method. Take E as constant.



Q. 6 Solve the following.

CO 12

5

- A) A Gas Cylinder of internal diameter 40mm is 5mm thick. If the tensile stress in the material is not to exceed 30Mpa, Find the maximum pressure which can be allowed in the cylinder. **6**
- B) A cylindrical vessel 2 m long and 500 mm in diameter with 10 mm thick plates is subjected to an internal pressure of 03 MPa. Calculate the change in the volume of the vessel. Take $E = 500\text{GPa}$ and poisson's ratio $= 0.3$ for the vessel material **6**

***** End of the paper *****